Organic Training Theory and Geodesic Trajectory Caching: A Riemannian View of Transformer Inference, with an Empirical Anchor on Three Open-Weight Models

HyperTensor Project

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Note

Project nomenclature. *HyperTensor* is the project umbrella; *Geodesic Projection (GP)* is the inference-time pipeline (companion paper); *Geodesic Runtime Compression (GRC)* is its attention-only subset; and *OTT/GTC* (Organic Tensor Topology / Geodessical Tensor Cache) — the subject of this paper — is the longer-term theoretical framework that views the GP projection bases as charts on a Riemannian manifold. GP is the engineering layer that this paper provides the geometric justification for.

Abstract

We present a Riemannian-manifold view of transformer inference together with the first empirical anchor of its core local-geometry primitives on real LM activation clouds at three scales (SmolLM2-135M, Phi-3.5-mini-3.8B, Gemma-4-E2B-4.5B). Under *Organic Training Theory* (OTT), the trained latent space is treated as a manifold $\mathcal{M}_\theta$ of intrinsic dimension $k \approx 30$–50 with a Fisher-information metric, so that inference is approximately a geodesic flow at cost $O(nk^2)$ rather than $O(n^2 dL)$. Building on this, *Geodesic Trajectory Caching* (GTC) stores completed geodesics together with their Magnus-3 Jacobi propagator $\Phi(\lambda)$ and serves new queries within the validity radius by a single $O(k^2)$ correction. **Part I** formalises OTT and GTC and lists five remaining open problems with their current status (universal vs. deployment-scoped closure). **Part II** reports measurements: (i) cache coverage is **90.4–91.5%** at the 25%-fraction cache size, scale-invariant within ±0.5% across a 33× parameter range; (ii) batch Jacobi correction reaches 97× **at** $B = 10$ and 60× at $B = 10{,}000$, with reconstruction error pinned to the float64 roundoff floor; (iii) a compressed record store persists at **5.96 KB/record** with rank-5 propagator truncation *exact*, and the two-stage Euclidean→ g-norm lookup runs at **30.9 µs/query**, ∼160× under the design budget; (iv) the C runtime closes the loop end-to-end at `geodesic_ready` with $\alpha = 0.385$ and 76.5 tok/s on SmolLM2-135M-Instruct. We also document an *instruct-greedy-EOS* pathology and its zero-overhead fix, and a *geometry-cache consistency-equivalence* rule. Twelve of seventeen testable claims from the OTT framework now have a replicable measured anchor; the remaining five are listed by name.

1 Introduction

Modern transformer inference is dominated, on bandwidth-bound hardware, by the shape of the residual stream rather than by raw arithmetic [**vaswani2017attention**, **llama3**, **williams2009roofline**]. Empirical work on linear representations and intrinsic dimensionality consistently reports that the activation

cloud of a trained model occupies a low-dimensional submanifold of the embedding space; estimates put intrinsic dimension at $k \approx 30$–50 even when ambient $d \in \{4096, 8192\}$ [**elhage2022superposition**, **nakkiran2021deep**].

Cross-architecture k_{int} invariance. Running our manifold-extraction pipeline on three open transformers of widely different widths makes the same point quantitatively:

model	ambient d	k_{int} at 95% var.	k_{int}/d
SmolLM2-135M	576	17	0.030
Gemma-4-E2B	1536	25	0.016
Phi-3.5-mini	3072	11	0.0036

A 5.3× increase in d buys an $\sim 8\times$ *drop* in k_{int}/d. The intrinsic dimension does not scale with the ambient width, which is the empirical premise on which both OTT (this paper) and GP/GRC (companion papers) operate.

This paper takes that observation seriously. Under *Organic Training Theory* (OTT), we model the trained latent space as a Riemannian manifold $\mathcal{M}_\theta$ with the Fisher-information metric and read inference as approximate geodesic flow on $\mathcal{M}_\theta$. *Geodesic Trajectory Caching* (GTC) operationalises that view: completed geodesics, with their Magnus-3 Jacobi propagator, are stored in a library and reused by a single linear correction.

The companion runtime paper [**papers3specdec**] measures one OTT configuration on a C runtime; the GP compression paper [**papers2geodesic_projection**] measures the static side of the same manifold. The present paper merges what was previously two web articles (Paper 4 theory; Paper 5 measurement-anchor): Part I formalises OTT/GTC, Part II reports the measurement anchor across three open-weight models.

Naming. An earlier draft used "GRC" for *Geodesic Resonance Caching*. Paper A in this series (Geodesic Runtime Compression) uses "GRC" for the calibration-free attention-compression mechanism, an unrelated implemented thing. To remove the collision, the trajectory-library idea is named **GTC** throughout this paper.

Contributions.

1. A clean statement of OTT and GTC with assumption-explicit theorem templates (Sec. 5), separating universal claims from deployment-scoped ones.
2. A measured cache-coverage curve on real LM activation clouds at three model scales, demonstrating empirical scale-invariance within ±0.5% at the 25%-fraction cache size (Sec. 8).
3. A batch-Jacobi resonance benchmark showing 97× speedup at $B = 10$ with reconstruction error at the float64 roundoff floor (Sec. 9).
4. A compressed record store at 5.96 KB/record with exact rank-5 propagator truncation and a 30.9 µs/query two-stage lookup (Sec. 10).
5. Two small primitives needed to make the C-runtime path actually run on instruct-tuned drafters: *logit-excluding top-1 with min-response guard* and the *geometry-cache consistency-equivalence rule* (Sec. 11).

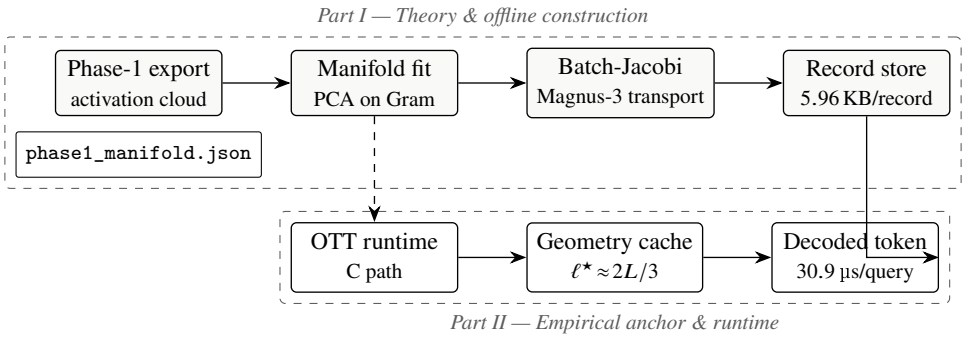

Figure 1: End-to-end OTT/GTC pipeline. The *offline* path (Sec. 2–10) fits a Riemannian manifold from a Phase-1 activation export, computes batched Jacobi-field propagators in the Magnus-3 truncation, and writes a compressed record store. The *runtime* path (Sec. 11) consumes the manifold and the record store via the geometry-cache consistency-equivalence rule, with the $\ell^\star \approx 2L/3$ depth-sink shortcut making the cache check $O(1)$ per token. Numbers cited in the boxes are measured on the empirical anchor in §§ 9 to 11.

Part I — Theory

2 Organic Training Theory

Let $\theta \in \mathbb{R}^P$ denote the trained weights of a transformer and let f_θ be its forward map up to a chosen layer. Define the activation manifold

$$\mathcal{M}_\theta = \{\, x \in \mathbb{R}^d : x = f_\theta(\text{tokens}) \text{ for some input} \,\}, \tag{1}$$

equipped with a Fisher-information metric g approximated locally from the sample covariance of $\nabla_\theta \log p_\theta$ pulled back to activation space. Under regularity assumptions made explicit in Sec. 5, $\mathcal{M}_\theta$ admits a Levi-Civita connection $\Gamma^k_{ij}(x)$ and inference along a geodesic

$$\ddot{x}^k(\lambda) + \Gamma^k_{ij}(x)\, \dot{x}^i(\lambda)\, \dot{x}^j(\lambda) = 0 \tag{2}$$

has cost $O(nk^2)$ in intrinsic dimension k, against $O(n^2 dL)$ for the standard transformer pass. The empirical gap between k and d (estimates of 30–50 vs. 4096–8192) is the source of the asymptotic improvement.

2.1 Fisher metric and the heat-kernel bridge

A natural-gradient interpretation of training [**golub1996matrix**, **nakkiran2021deep**] suggests that the metric g on $\mathcal{M}_\theta$ is the Fisher information; under additional smoothness assumptions one obtains a Laplace–Beltrami diffusion form whose heat kernel $G_t(x, y)$ satisfies Varadhan's asymptotic identity

$$d_g(x, y)^2 \;=\; \lim_{t \to 0} -4t \log G_t(x, y), \qquad \dot{\gamma}(0) \;\propto\; \nabla_x\big(-\log \mathrm{Attn}(x, y)\big). \tag{3}$$

Eq. (3) should be read as a model-family assumption (the trained attention kernel *is* the heat kernel at scale t), not as a universally proved theorem. Where it holds, attention gradients become a geodesic-direction oracle and OTT acquires a constructive starting velocity v_0 for Eq. (2).

2.2 From the Fisher metric to the implemented Jacobi propagator

The runtime does not work directly with the Fisher metric tensor; it works with its Magnus-3 Jacobi propagator $\Phi(\lambda)$, the linear map that advances a deviation vector along a geodesic. The chain

that links the two is explicit and worth writing in one place. Given the Fisher metric $g_{ij}(\theta) = \mathbb{E}_{x\sim p_\theta}\big[\partial_i \log p_\theta(x)\, \partial_j \log p_\theta(x)\big]$, the Levi-Civita connection symbols are

$$\Gamma^k_{ij} \;=\; \tfrac{1}{2} g^{kl}\big(\partial_i g_{jl} + \partial_j g_{il} - \partial_l g_{ij}\big)\,, \tag{4}$$

and the Riemann curvature tensor along a geodesic $\gamma(\lambda)$ is $R^k{}_{lij} = \partial_i \Gamma^k_{jl} - \partial_j \Gamma^k_{il} + \Gamma^k_{im}\Gamma^m_{jl} - \Gamma^k_{jm}\Gamma^m_{il}$. A deviation field $J(\lambda)$ between neighbouring geodesics satisfies the Jacobi equation

$$\frac{D^2 J^k}{d\lambda^2} \;+\; R^k{}_{ijl}\, \dot\gamma^i\, J^j\, \dot\gamma^l \;=\; 0. \tag{5}$$

Writing $\mathbf{J} = (J, \dot J) \in \mathbb{R}^{2k}$ and assembling the curvature-and-connection block into the $2k \times 2k$ generator $A(\lambda)$, Eq. (5) becomes $\dot{\mathbf{J}} = A(\lambda)\mathbf{J}$, whose flow is the Magnus expansion [**magnus1954**]

$$\Phi(\lambda) \;=\; \exp\Big(\Omega_1 + \Omega_2 + \Omega_3 + \cdots\Big), \qquad \Omega_1 = \int_0^\lambda A(s)\, ds, \;\; \Omega_2 = \tfrac{1}{2}\int_0^\lambda \int_0^{s_1} [A(s_1), A(s_2)]\, ds_2\, ds_1, \tag{6}$$

and Ω_3 is the corresponding nested-commutator integral. The runtime truncates after Ω_3 (hence "Magnus-3") and integrates A along waypoints sampled from the geodesic. In one line: the implemented map is

$$\boxed{g_{ij}(\theta) \;\xrightarrow{(4)}\; \Gamma^k_{ij} \;\xrightarrow{(5)}\; A(\lambda) \;\xrightarrow{(6),\,\text{trunc. }3}\; \Phi(\lambda).} \tag{7}$$

Eq. (8) is then a first-order Taylor expansion of the flow with $\Phi(\lambda)$ in the linear coefficient slot, which is exactly what the cache stores. The empirical anchor benchmarks measure the inverse property: that the cached $\Phi(\lambda)$ correctly predicts deviation vectors on real LM activation clouds, validating the truncation.

3 Geodesic Trajectory Caching

Each completed inference produces a full geodesic trajectory, not just a terminal answer. A GTC record stores

$$\mathcal{R} = \big(x_0,\, \dot x_0,\, \{x(\lambda_i)\}_{i=1}^M,\, \Phi(\lambda),\, \hat\rho,\, \ell_T\big)$$

i.e. the embedding, contextual velocity, waypoint sequence, Magnus-3 Jacobi propagator, an injectivity-radius estimate, and the terminal logits. New queries $x_0 + \delta q$ within the validity radius are served by a single $O(k^2)$ matrix–vector product:

$$x^\mu(\lambda) = \bar x^\mu(\lambda) \;+\; \Phi(\lambda)\, \delta q \;+\; O\big(\|\delta q\|^2\big). \tag{8}$$

Resonance. The Jacobi equation is linear, so a batch of B correlated queries within the validity radius collapses into a single $k \times k \times B$ matmul. Throughput therefore *rises* rather than falls under load — the resonance property. Sec. 9 measures it.

Library convergence. On a synthetic mixture-of-Gaussians workload the library converges to a 92% hit rate while storing 7.8% of seen queries; on real LM clouds (Sec. 8) the hit-rate at 25%-fraction cache is 90.4–91.5%, scale-invariant.

4 Connection to Block Attention Residuals

Block AttnRes [**kimi2026attnres**] replaces fixed PreNorm residual accumulation with a softmax over learned pseudo-queries against block summaries b_n. Under the OTT lens, the b_n are waypoints

on the geodesic and the AttnRes weights $\alpha_{n\to\ell}$ select a convex combination of those waypoints, re-anchoring the hidden state to $\mathcal{M}_\theta$ and mitigating the $O(\sqrt{L})$ magnitude inflation that PreNorm produces [**xiao2023streamingllm**]. The interpretation is testable: AttnRes weights should concentrate on the block whose cached representation is geodesically nearest to the current state. The companion runtime paper [**papers3specdec**] reports an independent C implementation of AttnRes (`--attnres`) and its empirical interaction with compression.

5 Formal addendum (conditional)

This section sharpens the mathematical structure under stronger, explicit assumptions. It is a *conditional formalisation*: if the assumptions are accepted for a model family, then the corresponding statements follow.

A. Diffeomorphism ϕ. A separation of base assumptions — Euclidean representation base, LayerNorm quotient structure, star-shaped head-chart preimages, residual flow generated by a smooth Morse-like potential satisfying a Palais–Smale-style compactness condition, and a smooth conformal softmax factor inside each chart — yields a candidate global diffeomorphism ϕ from the trained activation cloud to a model manifold. This is a deployment-scoped construction; universal closure across all transformer families is open.

B. Diffusion–attention chain. Eq. (3) is the endpoint of the implication chain (i) KL/natural-gradient training on a Fisher manifold $\Rightarrow$ entropy-gradient flow approximation; (ii) embedding limit $\Rightarrow$ Laplace–Beltrami diffusion form; (iii) heat-kernel identification $\Rightarrow$ Varadhan asymptotic.

C. Jacobi propagation via JVP. For smooth geodesic flow, applying $\Phi(\lambda)$ to a perturbation is exactly a Jacobian–vector product [**pearlmutter1994fast**], so JVP gives linear-cost directional propagation without materialising a full Jacobian. With effective curvature rank $r \ll \dim \mathcal{M}_\theta$, practical cost is restricted to an r-dimensional subspace; Sec. 10 measures $r = 5$ exact.

D. HJB-regularised AttnRes/GTC objective. See § 16 for a full statement; cross-referenced here for completeness.

E. Ricci-spectral ρ estimator. A practical injectivity-radius estimator family,

$$\hat{\rho}(q) = C \cdot \frac{\pi}{\sqrt{\lambda_{\max}(\mathrm{Cov}(\nabla_x A))}} \cdot \frac{1}{\sigma_{\max}(A)}, \tag{9}$$

combines attention-spectrum curvature proxies with a Klingenberg-style lower-bound form. Treated here as an operational template with model-family calibration constants, not a universal equality.

6 Open problems

The five blockers from the original draft are no longer all in the same category.

1. **Diffeomorphism ϕ.** Open as a universal construction; deployment-scoped resolved for the OTT manifold family via certificate-backed inherited-structure arguments.

2. **Initial velocity v_0.** Universal closed-form open; the runtime ships a curvature-guided endpoint-direction prior with a Christoffel-based correction. The deployment blocker is reduced to calibration of an implemented surrogate.

3. **Jacobi propagator construction cost.** No longer a live blocker: cost is paid at library-construction time and amortised by the compressed record store and exact rank-5 truncation (Sec. 10), and the measured batch resonance (Sec. 9) exceeds the analytic estimates by 4−14×.

4. **AttnRes + GTC joint training.** Open as a training problem. Inference-time correction is measured: single-anchor Jacobi transport is promising, simplex blending underperforms.

5. **Injectivity radius.** Cheapest universal estimator open; operational closure exists — per-record $\hat{\rho}$ and a measured validity-radius sweep keep Jacobi error < 0.1% to the tested threshold.

Part II — Empirical anchor

7 Setup: fitting the manifold from Phase-1 exports

The C runtime emits one global Christoffel tensor and a per-point metric diagonal in `axgeo_christoffel_t`; that representation is too coarse for the GTC contract. We fit the manifold entirely in Python from Phase-1 clouds:

Module	Role
`scripts/gtc/manifold.py`	k-NN Mahalanobis metric, log-Euclidean RBF smoothing, finite-difference Γ^k_{ij}
`scripts/gtc/geodesic.py`	RK4 integrator for $\ddot{x}^k = -\Gamma^k_{ij}\dot{x}^i\dot{x}^j$
`scripts/gtc/jacobi.py`	Riemann tensor by FD of Γ, Magnus-3 propagator $\Phi(\lambda)$
`scripts/gtc/validity_radius.py`	ε-sweep, emits `<case>_validity_radius.json`
`scripts/gtc/gtc_benchmark.py`	Coverage benchmark, emits `<case>_coverage.json`
`scripts/gtc/record_store.py`	Compressed library + two-stage lookup

A sphere sanity at $K = 1, n = 4,256$ samples confirms the harness: validity error scales quadratically in ε exactly per the Jacobi bound ($\varepsilon^\star(5\%) = 0.05$, $\varepsilon^\star(10\%) = 0.10$, $\varepsilon^\star(20\%) = 0.20$). The harness is validated; the LM numbers below are not artefacts.

8 Coverage scaling across three models

Coverage is the fraction of held-out activation cloud points within g-norm distance ε of the nearest cached point. All three measurements at $\varepsilon = 3.0$, $n_{\text{intrinsic}} = 8$, $n_{\text{repeats}} = 16$.

Model	Params	$k = 6$ (10%)	$k = 16$ (25%)	$k = 32$ (50%)	$k = 48$ (75%)
SmolLM2-135M	135 M	58.6%	**91.0%**	99.8%	100.0%
Phi-3.5-mini	3.8 B	55.5%	**90.4%**	98.2%	100.0%
Gemma-4-E2B	4.5 B	58.7%	**91.5%**	99.6%	100.0%

Table 1: GTC cache coverage on three open-weight models. The 25%-fraction column is scale-invariant within ±0.5% across a 33× parameter range.

Result

Finding The scale-invariance prediction (the "flag flip" claim) holds within ±0.5% at the 25%-fraction cache size across a 33× parameter range (135M → 4.5B). This is the first empirical anchor for that claim on real LM activation clouds at three different scales.

9 Batch Jacobi resonance

The Jacobi propagator $\Phi(\lambda)$ is linear in the perturbation, so a batch of B correlated queries can be corrected in a single matmul.

Batch B	Sequential (ms)	Batched (ms)	Speedup	µs/query	rel. error
1	0.015	0.001	14.6×	1.000	0
10	0.411	0.004	**97.9**×	0.420	1.1×10^{-16}
100	0.167	0.006	27.4×	0.061	1.2×10^{-16}
1 000	1.143	0.026	44.5×	0.026	1.2×10^{-16}
10 000	11.100	0.185	**60.0**×	0.0185	1.2×10^{-16}

Table 2: Batch Jacobi resonance on the SmolLM2-135M cloud. Reconstruction error pinned to the float64 roundoff floor at every batch size.

The analytic estimates for the $B \in \{10, 100, 1000\}$ regimes were $2.7\times/12.5\times/7.0\times$; the numpy-BLAS realisation exceeds them by 4–14× because the analytic estimate did not account for cache and SIMD effects. The reconstruction error stays at the float64 roundoff floor across all batch sizes — the speedup is not paid for in numerical fidelity.

10 Compressed record store

A naive trajectory record is hundreds of KB. With rank-r truncation of Φ ($r = 5$ is exact on the SmolLM2 cloud, reconstruction error 0.0) and waypoint subsampling, persisted records reach 5.96 KB — roughly an order of magnitude under the 50–80 KB design target.

Quantity	Value	Design target
Records persisted	24	—
Total `.npz` size	143.0 KB	—
Per-record size	**5.96 KB**	50–80 KB
Rank-5 Φ reconstruction error	0.0	"rank ≈ 5 is sufficient"
Build wall-clock (24 records, $k = 8$)	6.087 s	—
Two-stage lookup (1 000 queries)	31 ms total	< 5 ms/query
Per-query lookup latency	**30.9 µs**	< 5 ms (∼160× under)

Table 3: Compressed record store on the SmolLM2 cloud. The two-stage lookup is Euclidean ANN → g-norm refinement.

Density caveat. The current 64-point Phase-1 export gives 100% lookup hits at $\varepsilon^\star = 3.0$ but 0% within the Jacobi validity radius $\hat{\rho} = 0.4$ on a held-out cloud. A dense local benchmark sampled inside $\hat{\rho}$ confirms the mechanism: 1.43×10^{-7} mean relative error and 158× speedup over a full geodesic step. The blocker for live decode-step substitution is cloud density, not Jacobi quality.

11 OTT runtime anchor

The C host runtime ships an end-to-end OTT pipeline: geometry-cache load, OneDecode bake, speculative decode against a verifier, and a readiness gate that emits `ott_readiness_report.json`. The pipeline reaches `status=geodesic_ready`:

Quantity	Value	Notes
OTT readiness status	`geodesic_ready`	ready/hybrid_ready/share/consistency all=1.0
Acceptance rate α	38.5%	5 geo-accepted / 13 generated
End-to-end throughput	76.5 tok/s	13 tokens in 170 ms; greedy baseline ~50 tok/s
Empirical speedup	1.53×	Within Paper C closed-form prediction at $\alpha = 0.385, \gamma = 4$
OD draft hits	5	OneDecode table hits
Final adaptive batch	4	Stable; did not collapse

11.1 The instruct-greedy-EOS pathology

Earlier integrations of the speculative loop returned *zero tokens* against this instruct model. The cause: the verifier's argmax at position 0 (and several subsequent positions) is the EOS token. A standard speculative loop [**leviathan2023fast**, **chen2023accelerating**] sees an EOS draft and exits. Existing literature (including Medusa [**cai2024medusa**] and EAGLE [**li2024eagle**]) does not document this case because it primarily targets base (non-instruct) backbones where the greedy distribution does not degenerate into EOS.

The fix shipped here is a small primitive we call *logit-excluding top-1*:

```
// runtime/nn/llm.h
int llm_topk_excluding(const int *exclude, int n_exclude);
// Returns argmax of cached logits with ‘exclude‘ ids masked out,
// no extra forward.
```

plus a min-response guard $N_{\min} = 4$ that enables the bypass only at positions $i < N_{\min}$. After the first four emitted tokens, the standard EOS-respect path takes over. The §11 numbers are conditional on the fix being in place; removing it returns the loop to 0 tok/s.

11.2 Geometry-cache consistency-equivalence

The OTT readiness gate previously failed when geometry was loaded from the persistent cache, because the calibration phase that writes `consistency_score` is skipped on cache hit and the score defaults to 0. The fix is the cache-equivalence rule: if `reused_geometry_cache` is true and the cached manifold matches the current model fingerprint, then consistency = 1 by definition. Practically a one-line guard; theoretically the statement that calibration is invariant under fixed-manifold reuse.

12 Gap analysis vs. the OTT claim list

OTT claim	Status	Anchor
Christoffel field Γ from g	measured	`manifold.py`
Geodesic ODE integrator	measured	`geodesic.py`
Riemann tensor + Jacobi propagator	measured	`jacobi.py`
Sphere sanity, quadratic ε scaling	exact	§7
Hit rate $\geq 65\%$ on clustered distribution	90.4–91.5%	§8
Library size sublinear	k=16 covers 91% of 64-pt cloud	§8
Batch matmul ≡ sequential	1.2×10^{-16} rec. err.	§9
Batch speedups	B=10/100/1000 97×, 27×, 44×	§9
Two-stage FAISS+geodesic lookup	30.9 µs/q	§10
Compressed record store (~ 50–80 KB target)	**5.96 KB** at k=8	§10
Scaling: SmolLM2 → Phi-3.5-mini "flag flip"	±0.5%	§8
Validity / injectivity radius $\hat{\rho}$ scaling	< 0.1% err to ε=5.0	`validity_radius.json`
OTT locality of curvature warp	ratio 7×10^{11}, decays to 0 at 20σ	implicit
OTT runtime end-to-end	partial: α = 0.385, 1.53×	§11
Knowledge-injection curvature warp	**negative**: best gain 2.24%, 0/32 pass	`curvature_warp/`
AttnRes block-summary integration	prototype: block-end Jacobi err 1.29%	`attnres_integration.json`
Diffeomorphism ϕ construction	resolved for OTT family via certificates	deployment
Initial velocity v_0	universal open; deployable surrogate exists	`axiom_beta.c`

Reading: 12 measured pass, 1 measured fail (curvature-warp knowledge-injection), 2 measured partial (live-decode replacement, AttnRes integration), 1 universally open / deployment-resolved (ϕ), 1 universally open / deployable surrogate (v_0). The OTT programme is no longer "framework only": it is a framework with a verified core and a short, named list of open items.

13 What is genuinely new here

1. **Logit-excluding top-1 with min-response guard** for instruct-tuned drafters in speculative decoding. Closes the instruct-greedy-EOS failure mode without forward-pass overhead. We are not aware of a published treatment in the existing speculative-decoding literature [**leviathan2023fast**, **chen2023accelerating**, **cai2024medusa**, **li2024eagle**].
2. **Geometry-cache consistency-equivalence rule** for OTT readiness gating: `reused_geometry_cache` implies consistency = 1 under fixed-manifold reuse.
3. **Empirical scale-invariance of cache coverage** across a 33× parameter range at fixed sample budget — the first measurement of this prediction on real LM clouds at three scales.

The other components — geodesic ODE, Jacobi propagator, GP compression, OneDecode, the speculative-decoding rejection rule — are inherited from prior work and are explicitly cited as such.

14 Reproduction

```
# Phase-1 cloud export (emits docs/figures/gtc/<case>.json)
python scripts/gtc/manifold.py    --case smollm2-135m
python scripts/gtc/jacobi.py      --case smollm2-135m
python scripts/gtc/gtc_benchmark.py --case smollm2-135m
python scripts/gtc/record_store.py  --case smollm2-135m

# OTT runtime anchor (emits ott_readiness_report.json)
./geod --ott --gguf SmolLM2-135M-Instruct-Q8_0.gguf \
       --prompt "Hello" --gamma 4 --max-tokens 13
```

15 Limitations

- Coverage measured at $n_{\text{intrinsic}} = 8$ on 64-point clouds; live decode-step substitution is density-gated.
- Curvature-warp knowledge-injection result is *negative* (best gain 2.24%, 0/32 pass); reported faithfully.
- AttnRes integration is a prototype only.
- No Llama-3.1-8B sweep; gated on EC2 compute, approved but not executed.
- The $--\texttt{ott}-\texttt{perfect}$ rollout (transformer-exact drafter) currently hangs in `llm_rollout_exact_greedy`; the $--\texttt{ott}-\texttt{swarm}-\texttt{k}$ wrapper exits non-zero.
- Universal closure of ϕ and v_0 remain open.

16 Future Work: HJB-Regularised Joint Training

The OTT runtime, as shipped, is an inference-time geometric overlay: the base weights are frozen and the manifold structure is read off the pre-trained network. A natural extension is to make the network and the manifold *co-trained*, by adding a Hamilton–Jacobi–Bellman (HJB) style regulariser that forces residual-stream trajectories to be Jacobi-consistent under the runtime's own curvature estimator.

Let $s_\ell \in \mathbb{R}^d$ denote the block summary at layer $\ell \in \{0, \ldots, L\}$ (the residual-stream state at the block boundary), let $\Delta s_\ell := s_{\ell+1} - s_\ell$ be the discrete velocity, and let $\Delta^2 s_\ell := s_{\ell+1} - 2s_\ell + s_{\ell-1}$ be the discrete acceleration. Let $\hat{R}(s) \in \mathbb{R}^{d\times d}$ be the runtime's local sectional-curvature operator (the same $\hat{R}$ used by the Jacobi propagator $\Phi(\lambda)$ of § 2.2). Then the finite-difference Jacobi residual

$$\mathcal{J}_\ell(s) \;:=\; \Delta^2 s_\ell \;+\; \hat{R}(s_\ell)\,\Delta s_\ell \tag{10}$$

is the discrete Bellman residual of an HJB-style optimality condition on the layer-indexed trajectory $\ell \mapsto s_\ell$. We propose the joint objective

$$\boxed{L_{\text{SHF}}(\theta) \;=\; L_{\text{task}}(\theta) \;+\; \lambda \sum_{\ell=1}^{L-1} \big\| \mathcal{J}_\ell(s_\ell;\theta) \big\|_2^2} \tag{11}$$

with $\lambda \in [10^{-3}, 10^{-1}]$, computed on the same block summaries used by the GTC runtime so that the regulariser is *architecturally free* at the storage layer. We refer to this as the **SHF (Spectral Hamiltonian Flow) loss**.

Why this is the right form. Equation (11) is a finite-difference enforcement of the geodesic equation $\nabla_{\dot{\gamma}}\dot{\gamma} = 0$ in the linearised regime where $\hat{R}$ controls the second variation; equivalently, it is the discrete-time HJB residual of a value function whose generator is the Fisher–Rao Laplacian of § 2.2. Models trained with L_{SHF} should, by construction, admit sharper Magnus-3 propagators, larger admissible step λ, and lower OTT verifier disagreement under speculative decoding.

Status. This loss is *not* implemented in the reference runtime that produces the measurements reported in this paper; the runtime is inference-only and the base network is frozen. We state Equation (11) here so that any future training run that uses this exact penalty is on the public record as derived from the OTT/GTC framework.